

DIPOLE TYPE BY LENGTH
l = antenna rod length (m)

λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
<i>l</i> /λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
= 1	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors <i>small l</i> ≪ λ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$ Avg power density <i>l</i>/λ < 1/50; far field	
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta$ (W/m ²)	
<i>Equivalent:</i> $S = S_{\text{max}} \sin^2 \theta$ with $S_{\text{max}} = \eta_0 k^2 I_0^2 l^2 / (32\pi^2 R^2)$ at $\theta = \pi/2$.	
<i>l</i> : rod length (m) <i>R</i> : obs. distance (m) <i>k</i> = 2π/λ (rad/m)	<i>I</i> ₀ : peak current (A) <i>θ</i> : angle from dipole axis (rad) <i>η</i> ₀ = 120π ≈ 377 Ω
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l / \lambda)^2$ (W)	

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Avg power density any <i>l</i>; far field	
$S(R, \theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$ (W/m ²)	
<i>At θ = 0, π: bracket is 0/0. L'Hôpital ⇒ S = 0 (no axial radiation). TI-36X alt: evaluate bracket at θ = 0.001 rad → tiny → 0.</i>	
Normalized pattern dimensionless $F(\theta) = S(\theta) / S_{\text{max}}$	

COMMON ANGLES (deg ↔ rad, trig values)

Conversion multiply by π/180 or 180/π $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$						
°	rad	sin θ	cos θ	sin ² θ	cos ² θ	θ
0	0	0	1	0	1	
30	π/6	1/2	√3/2	1/4	3/4	
45	π/4	√2/2	√2/2	1/2	1/2	
60	π/3	√3/2	1/2	3/4	1/4	
90	π/2	1	0	1	0	
180	π	0	−1	0	1	

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$ $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$	
Normalize $F_a^{\text{norm}} = F_a / N^2$	
HPBW (broadside, uniform) use <i>Nd</i>, not (<i>N</i>−1)<i>d</i> $\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = 50.8° · $\frac{\lambda}{Nd}$	
Electronic steering (scan to θ ₀) replace γ with γ' = $kd(\cos \theta - \cos \theta_0)$ $\delta = \frac{2\pi d}{\lambda} \cos \theta_0$ ($\frac{\text{rad}}{\text{elr}}$); $\theta_0 = \frac{\pi}{2}$: broadside, 0: end-fire	
Frequency scan feed-line incr. Δ<i>l</i> = <i>n</i>₀ λ₀ $\cos \theta_0 \approx \frac{n_0 \lambda_0}{d} \frac{\Delta f}{f_0}$	
<i>Grating lobes appear if d > λ. Avoid.</i>	

BEAM PARAMETERS — β, Ω_p, *D*

Normalize always start here $F(\theta) = S(\theta) / S_{\text{max}}$ (unitless) Beamwidth β at threshold <i>X</i> Set $F(\theta_X) = X$, solve for θ _X ; β = 2θ _X (symmetric)		
Threshold	$X = 10^{\text{dB}/10}$	θ _X
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any <i>X</i>	$F(\theta_X) = X$
Pattern solid angle Ω _p recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr) <i>No φ dep.: Ω_p = 2π ∫₀^π F(θ) sin θ dθ.</i> <i>θ is the integration variable — DON'T plug in a number for θ (e.g. θ_{1/2}). Integrate over θ ∈ [0, π].</i>		
Pattern $F(\theta)$	Ω _p (sr)	
Isotropic ($F=1$)	$4\pi \approx 12.57$	
Hertzian sin ² θ	$8\pi/3 \approx 8.38$	
Half-wave dipole	≈ 7.66	
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2 / \ln 2$	
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$	
Other (use integral)	numerical	

TI-36X: [2nd] [d/dx] for $\int \square$, RAD mode, multiply by 2π.

Directivity <i>D</i> always $D = 4\pi/\Omega_p$; common shortcuts below $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$		
WORD TRAP: “direction” = θ _{max} (rad); “directivity” = <i>D</i> (unitless).		
Gain (if asked) ξ = radiation efficiency (unitless), 0 < ξ ≤ 1 $G = \xi D$ (unitless) $G_{\text{dB}} = 10 \log_{10} G$ (dB)		
Lossless / ideal antenna ⇒ <i>G</i> = <i>D</i> .		
$\xi = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}$; $R_{\text{rad}} = 2P_{\text{rad}}/I_0^2$ (Ω); $R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$.		

COMMON DIPOLE PARAMETERS

Lossless (ξ = 1): <i>G</i> = <i>D</i> .				
Type	<i>l</i>	<i>D</i>	<i>D</i> _{dB}	<i>R</i> _{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	< λ/50	1.5	1.76	80π ² (<i>l</i> /λ) ²
Half-wave	λ/2	1.64	2.15	73.2
Monopole	λ/4 (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199
Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole (λ/4 above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.				

ANTENNA AREAS (*A_e*, *A_p*)

Effective area antenna's RX cross section; <i>D</i> from BEAM PARAMS or table above $A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section wire dipole, diameter <i>d</i>; <i>l</i> see table above $A_p = l d$ (m ² , any dipole)	
Inverse: <i>D</i> from <i>A_e</i> $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
<i>Thin wire:</i> $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).	

FRISIS TRANSMISSION FORMULA

<i>For each antenna's <i>G</i>: if given by NAME (“half-wave dipole” etc.) → COMMON DIPOLE PARAMS (<i>G</i> = <i>D</i>, lossless). Else (dB/linear value given) → use as given (convert dB → linear).</i>	
Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$ $S_r = \frac{G_t P_t}{4\pi R^2}$ (W/m ²); $P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$ (W)	
Equivalent: $P_r = S_r A_{e,r}$. Solve for <i>R</i> range $R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$ (m)	
Equivalent form using A_e recv. area form $P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$, $G_r = \frac{4\pi A_{e,r}}{\lambda^2}$	
Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.	
Off-axis (patterns not aligned): multiply by $F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$, both = 1 when aligned.	
Recipe 1. λ = <i>c</i> / <i>f</i> . 2. Convert any dB gain → linear: $G = 10^{G_{\text{dB}}/10}$.	
3. Get <i>G_t</i> , <i>G_r</i> by antenna type: dipole→COMMON DIPOLE PARAMS; aperture→APERTURE box; arbitrary <i>F</i> →BEAM PARAMS. 4. Plug into Friis.	
<i>Use linear <i>G</i>, not dB. Convert FIRST.</i>	

dB ↔ LINEAR

Works for any power-like quantity (<i>G</i> , <i>D</i> , <i>P_r</i> / <i>P_t</i> , <i>SNR</i> , ...): $X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$ $X_{\text{lin}} = 10^{X_{\text{dB}}/10}$ Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.	
<i>For E-field / amplitude use 20 log not 10 log (power ∝ E²).</i>	

NOISE & SNR

Noise power thermal noise into matched receiver $P_N = k_B T_{\text{sys}} B$ (W)	
$k_B = 1.38 \times 10^{-23}$ J/K; <i>T_{sys}</i> system noise temp (K); <i>B</i> receiver bandwidth (Hz).	
Signal-to-noise ratio $\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$	

APERTURE ANTENNAS (horn, dish)

<i>A_p</i> physical aperture area (m ²); <i>A_e</i> effective area (m ² , = λ ² <i>D</i> /4π); β _{xz} , β _{yz} HPBW's in two planes (rad); <i>ℓ</i> , <i>d</i> aperture side / diameter (m). <i>Far-field req.: R ≥ 2d²/λ (d = largest aperture dim.).</i>			
Pattern (rect., uniform illum.) peak at θ = 0 (broadside) $F(\theta) = \text{sinc}^2\left(\frac{\pi \ell_x \sin \theta}{\lambda}\right)$; $S_0 = \frac{E_0^2 A_p^2}{2\eta_0 \lambda^2 R^2}$			
Directivity (any shape, uniform illum.) $D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) ⇒ $A_e \approx A_p$			
Directivity from beamwidths any aperture $D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$			
HPBW and <i>D</i> by aperture shape uniform illum.; β in rad			
Shape	<i>A_p</i>	HPBW (rad)	<i>D</i> (unitless)
Rect. $\ell_x \times \ell_y$		$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. <i>d</i>)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2 / \lambda^2$
Scaling (any ratio) $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$ $D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}}\right)^2$ $\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$ <i>If a parameter is unchanged, its ratio = 1. Examples: double area → D×2, β×1/√2; double freq → D×4, β×1/2.</i>			

DIPOLE

<i>l</i> — antenna length (m) λ = <i>c</i> / <i>f</i> (m); <i>c</i> = 3×10 ⁸ m/s <i>I</i> ₀ — peak current (A) <i>k</i> = 2π/λ (rad/m) <i>R</i> — obs. dist. (m); θ from axis <i>η</i> ₀ = 120π ≈ 377 Ω $S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless) <i>a</i> — coef. on θ ² in Gaussian β — HPBW (rad or °) Ω _p — pattern solid angle (sr) <i>D</i> = 4π/Ω _p — directivity ξ radiation eff. (unitless); <i>G</i> = ξ <i>D</i> $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$
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FRISIS / dB

<i>P_t</i> — power transmitted (W) <i>P_r</i> — power received (W) <i>G_t</i> — transmit gain (linear) <i>G_r</i> — receive gain (linear) $S_r = G_t P_t / (4\pi R^2)$ (W/m ²) $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$ $G = 10^{G_{\text{dB}}/10}$ 3 dB = ×2, 10 dB = ×10 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

<i>A_p</i> — physical area (m ²) $A_e = \lambda^2 D / (4\pi)$ (m ²) $A_e \approx A_p$ for aperture $D \approx 4\pi A_p / \lambda^2$ (unitless) β ≈ 0.88λ/ℓ (rect/sq.) β ≈ 1.02λ/ <i>d</i> (circular) $2A_p \Rightarrow 2D$, β/√2

ARRAY

<i>N</i> — number of elements <i>d</i> — spacing (m) γ = <i>k d</i> cos θ $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$ $F_a^{\text{max}} = N^2$ at θ = π/2 $F_a^{\text{norm}} = F_a / N^2$ $\beta \approx 0.88\lambda/(Nd)$ (broadside)
